

Technical Data Sheet

PVT – the solar panel with both photovoltaic and thermal energy production



Article No
PVT00205



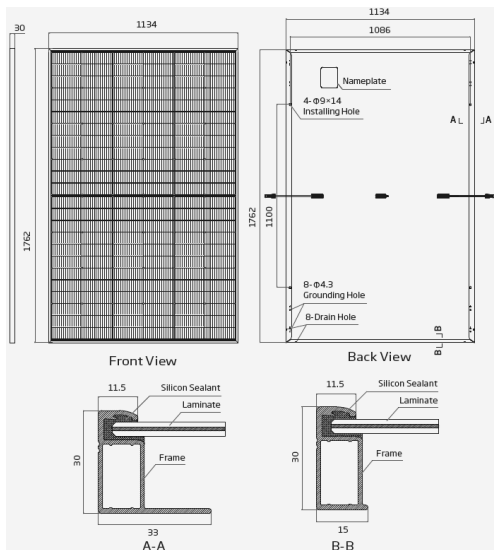
General description

PVT00205 is a PVT solar panel (photovoltaic thermal), containing an electric PV-panel and a thermal absorber combined.

The PVT solar panel is designed to lead away excess heat that is built up in the PV-panel from the solar irradiance and make the PV-panel work at its optimal temperature.

The thermal energy can be used in various applications and the electrical energy production will increase due to the optimization of the panel's operating temperature.

Dimensions



General data

Physical properties			
Dimensions	L x W x H	1762 x 1134 x 30	mm
Total Area	Gross	2,0	m ²
Weight	Empty Full	25,0 26,8	kg
Max Static Loading	Front Pull	5400 2400	Pa
Hailstone Test	25 mm Hailstone	at 23	m/s
Heat Strengthened Dual-glass	High Transmission, AR Coated	front glass 1,6	mm
		back glass 1,6	mm
Output Cable	4 mm ²	1200	mm
Number of Cells	mono-crystalline silicon cells	144 (6x24)	
Connection Box	Three diodes	IP68	
Frame	Black	Anodized aluminium alloy	

Photovoltaic data

Electrical characteristics (STC)			
Nominal Power	P _{max}	425 ~ 445	W _p
Maximum Power Voltage	U _{mpp}	42,9 ~ 44,3	V
Maximum Power Current	I _{mpp}	9,92 ~ 10,05	A
Open Circuit Voltage	V _{oc}	50,9 ~ 52,6	V
Short Circuit Current	I _{sc}	10,56 ~ 10,71	A
Module efficiency		21,3 ~ 22,3	%

Temperature coefficient (STC)			
Temp Coeff Max Panel Power	P _{mpp}	-0,30	%/°C
Temp Coeff Open Circuit Voltage	V _{oc}	-0,24	%/°C
Temp Coeff Short Circuit Current	I _{sc}	+0,04	%/°C

Operating parameters		
Operational Temperature	-40 ~ +85	°C
Max System Voltage (IEC/UL)	DC1500	V
Maximum Series Fuse Rating	25	A
NOCT	45 ± 2	°C
Protection class	Class II	
Fire Rating	UL type 1 or 2	IEC Class C

Standard test conditions STC: AM 1.5, Irradiation 1000 W/m², Cell temperature 25°C
Test uncertainty for P_{max}: +/- 3%

Thermal data

Collector characteristics			
Thermal Power Output	Peak	2750 W _{th} /panel	1430 W _{th} /m ²
Thermal Power Output	ΔT=0	1020 W _{th} /panel	530 W _{th} /m ²
Optical Efficiency	η ₀	0,53	
Heat Loss Coeff 1 st order	a ₁	22,5 W/m ² ·K	
Heat Loss Coeff 2 nd order	a ₂	0 W/m ² ·K ²	
Area	A _{gross} A _{aperture}	2,00 m ² 1,96 m ²	
Heat Transfer Material	plate pipe	Alu Alu	

Thermal Power Output Peak is calculated at T_m-T_a=-40 K, G=1000 W/m²

Heat transfer characteristics			
Thermal Fluid Volume		1,73 L	
Max Operation Pressure		15 bar	
Stagnation Temperature		80°C	
Hydraulic Connection		12 mm Pushfit	
Pressure Loss	at 120 L/h	8,6 kPa	0,83 mH ₂ O
	at 180 L/h	13,1 kPa	1,27 mH ₂ O

Pressure Loss is calculated at T=20°C, Heat Transfer Fluid: Water/Glykol 60/40

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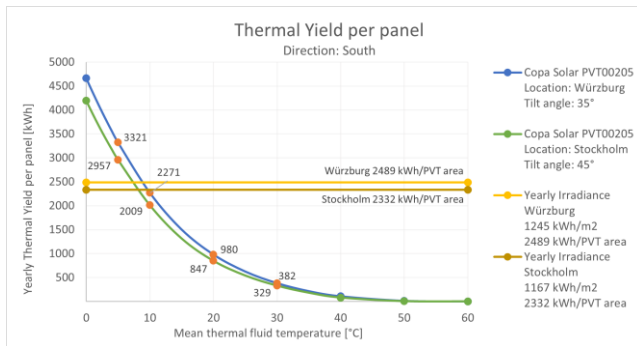


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Yearly Thermal Yield

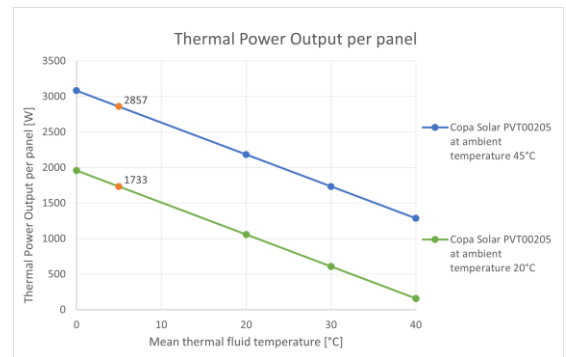
Calculation of Thermal Yield is made with ScenoCalc v6.2.



Thermal Power Output

Thermal Power Output is evaluated in relation to ambient temperature at 45°C and 20°C.

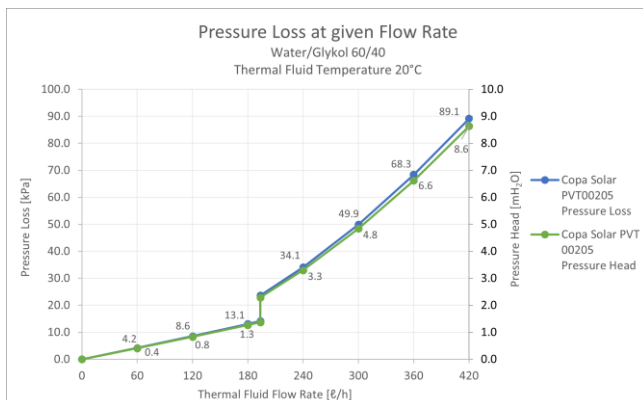
The thermal fluid is performing the best heat transport at low mean temperatures.



Pressure drop

The pressure drop and pressure head are calculated with the pipe characteristics.

The laminar–turbulent transition occurs at ~193 L/h.



Additional information

Conforms to the following standards

EN 12975:2022	SS EN 61215-1:2021
EN 62938	SS EN 61215-2:2021
EN 61730-1	SS EN 61215-1-1:2021
	SS EN 61215-1-2:2021
	SS EN 61215-1-3:2021
	SS EN 61215-1-4:2021

UN CPC code

171 Electrical energy
173 Steam and hot water

NACE Code

D351133 Alternative and renewable sources
of energy: solar

GPC code

10005875 Solar Power Stations

Model No

PVT-M-ALAL-1017-445A

Subject to technical modifications and any changes.

